



GyaanVault

SPPU M.Sc. Data Science — Free Study Material

Artificial Intelligence

DS-512-MJ

Sample / Practice Question Paper

Semester I · Elective · Practice Paper

Time: 3 Hours

Maximum Marks: 35

Instructions: (1) All questions carry equal weightage where indicated. (2) Figures to the right indicate full marks. (3) Draw neat diagrams / write code wherever necessary.

Q1. Answer any FIVE of the following. [5 × 2 = 10]

- a) What is an intelligent agent?
- b) Define state space search.
- c) What is a heuristic?
- d) Differentiate BFS and DFS.
- e) What is the A* algorithm?
- f) Define propositional logic.
- g) What is forward chaining?

Q2. Answer the following. [10]

A) Explain types of intelligent agents and their structure.

OR

B) Explain the Water Jug and 8-puzzle problems as state space search.

Q3. Answer the following. [10]

A) Explain uninformed and informed search algorithms (BFS, DFS, A*).

OR

B) Explain the A* and AO* search algorithms with examples.

Q4. Answer the following. [5]

A) Explain knowledge representation techniques.

OR

B) Explain propositional and predicate logic with forward/backward chaining.

